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BSP742RXUMA1 BSP742RIXUMA1

EN
This Datasheet is presented by the manufacturer

DE
Dieses Datenblatt wird vom Hersteller bereitgestellt

FR
Cette fiche technique est présentée par le fabricant

Smart Power High-Side-Switch

Features

- Overload protection
- Current limitation
- Short circuit protection
- Thermal shutdown with restart
- Overvoltage protection (including load dump)
- Fast demagnetization of inductive loads
- Reverse battery protection with external resistor
- Open drain diagnostic output
- Open load detection in OFF - State
- CMOS compatible input
- Loss of GND and loss of V_{bb} protection
- ESD - Protection
- Very low standby current

Product Summary

Overvoltage protection	$V_{bb(AZ)}$	41	V
Operating voltage	$V_{bb(on)}$	5...34	V
On-state resistance	R_{ON}	350	mΩ
Nominal load current	$I_{L(nom)}$	0.4	A



P-DSO-8

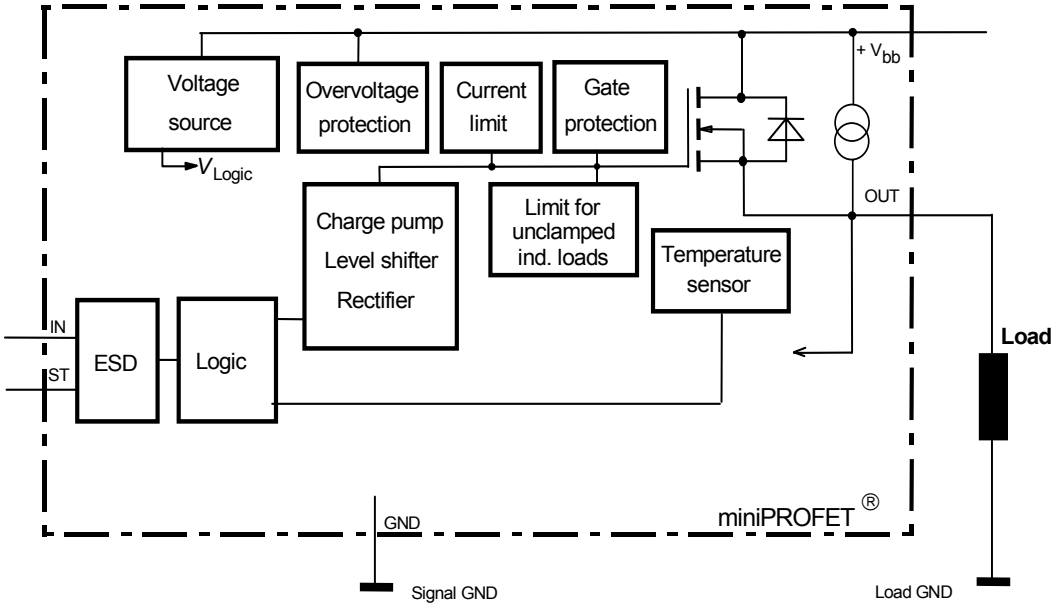
Application

- All types of resistive, inductive and capacitive loads
- µC compatible power switch for 12 V and 24 V DC applications
- Replaces electromechanical relays and discrete circuits

General Description

N channel vertical power FET with charge pump, ground referenced CMOS compatible input and diagnostic feedback, monolithically integrated in Smart SIPMOS® technology. Providing embedded protective functions.

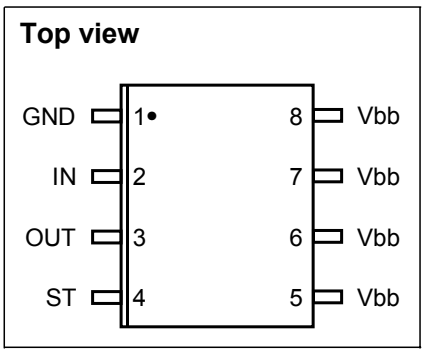
Block Diagram



Pin	Symbol	Function
1	GND	Logic ground
2	IN	Input, activates the power switch in case of logic high signal
3	OUT	Output to the load
4	ST	Diagnostic feedback
5	Vbb	Positive power supply voltage
6	Vbb	Positive power supply voltage
7	Vbb	Positive power supply voltage
8	Vbb	Positive power supply voltage

Pin configuration

Top view



Maximum Ratings at T_j = 25°C, unless otherwise specified

Parameter	Symbol	Value	Unit
Supply voltage	V_{bb}	40	V
Supply voltage for full short circuit protection	$V_{bb(SC)}$	V_{bb}	
Continuous input voltage	V_{IN}	-10 ... +16	
Load current (Short - circuit current, see page 5)	I_L	self limited	A
Current through input pin (DC)	I_{IN}	± 5	mA
Operating temperature	T_j	-40 ...+150	°C
Storage temperature	T_{stg}	-55 ... +150	
Power dissipation 1)	P_{tot}	1.5	W
Inductive load switch-off energy dissipation 1)2) single pulse, (see page 9) Tj =150 °C, V_{bb} = 13.5 V, I_L = 0.3 A	E_{AS}	800	mJ
Load dump protection 2) $V_{LoadDump}$ 3)= V_A + V_S R_I=2Ω, t_d=400ms, V_{IN}= low or high, V_A=13,5V R_L = 45 Ω	$V_{Loaddump}$	60	V
Electro**s**tatic **d**ischarge voltage (Human Body Model) according to ANSI EOS/ESD - S5.1 - 1993 ESD STM5.1 - 1998 Input pin all other pins	V_{ESD}	 ± 1 ± 5	kV

Thermal Characteristics

Thermal resistance @ min. footprint	$R_{th(JA)}$	-	95	-	K/W
Thermal resistance @ 6 cm² cooling area 1)	$R_{th(JA)}$	-	70	83	

1Device on 50mm*50mm*1.5mm epoxy PCB FR4 with 6 cm2 (one layer, 70µm thick) copper area for drain connection. PCB is vertical without blown air. (see page 17)

2not subject to production test, specified by design

3$V_{Loaddump}$ is setup without the DUT connected to the generator per ISO 7637-1 and DIN 40839 .

Supply voltages higher than $V_{bb(AZ)}$ require an external current limit for the GND pin, e.g. with a 150Ω resistor in GND connection. A resistor for the protection of the input is integrated.

Electrical Characteristics

Parameter and Conditions at T_j = -40...+150°C, V_{bb} = 13,5V, unless otherwise specified	Symbol	Values			Unit
		min.	typ.	max.	
Load Switching Capabilities and Characteristics					
On-state resistance T_j = 25 °C, I_L = 0.3 A, V_{bb} = 9...40 V T_j = 150 °C	R_{ON}	 - -	 250 450	 350 700	mΩ
Nominal load current Device on PCB [1)2)] T_C = 85 °C, $T_j \leq$ 150 °C	$I_{L(nom)}$	0.4	-	-	A
Turn-on time to 90% V_{OUT} R_L = 47 Ω, V_{IN} = 0 to 10 V	t_{on}	-	-	140	µs
Turn-off time to 10% V_{OUT} R_L = 47 Ω, V_{IN} = 10 to 0 V	t_{off}	-	-	170	
Slew rate on 10 to 30% V_{OUT}, R_L = 47 Ω	dV/dt_{on}	-	-	2	V/µs
Slew rate off 70 to 40% V_{OUT}, R_L = 47 Ω	$-dV/dt_{off}$	-	-	2	
Operating Parameters					
Operating voltage	$V_{bb(on)}$	5	-	34	V
Undervoltage shutdown of charge pump	$V_{bb(under)}$	-	-	5	
Undervoltage restart of charge pump	$V_{bb(u\ cp)}$	-	-	5.5	
Standby current V_{IN} = 0 V	$I_{bb(off)}$	-	-	26	µA
Leakage output current (included in $I_{bb(off)}$)	$I_{L(off)}$	-	-	12	
Operating current V_{IN} = 5 V	I_{GND}	-	-	1.3	mA

[1]Device on 50mm*50mm*1.5mm epoxy PCB FR4 with 6 cm2 (one layer, 70µm thick) copper area for drain connection. PCB is vertical without blown air. (see page 17)

[2]Nominal load current is limited by current limitation (see page 5)

Electrical Characteristics

Parameter and Conditions at T_j = -40...+150°C, V_{bb} = 13,5V, unless otherwise specified	Symbol	Values			Unit
		min.	typ.	max.	
Protection Functions[1)]					
Initial peak short circuit current limit (pin 5 to 3)	$I_{L(SCp)}$				A
T_j = -40 °C, V_{bb} = 20 V		-	-	2	
T_j = 25 °C		-	1.2	-	
T_j = 150 °C		0.4	-	-	
Repetitive short circuit current limit $T_j = T_{jt}$ (see timing diagrams)	$I_{L(SCr)}$	-	1	-	
Output clamp (inductive load switch off) at $V_{OUT} = V_{bb} - V_{ON(CL)}$, I_{bb} = 4 mA	$V_{ON(CL)}$	41	47	-	V
Overvoltage protection [2)] I_{bb} = 4 mA	$V_{bb(AZ)}$	41	-	-	
Thermal overload trip temperature	T_{jt}	150	-	-	°C
Thermal hysteresis	ΔT_{jt}	-	10	-	K
Reverse Battery					
Reverse battery[3)]	$-V_{bb}$	-	-	32	V
Drain-source diode voltage ($V_{OUT} > V_{bb}$) T_j = 150 °C	$-V_{ON}$	-	600	-	mV

[1] Integrated protection functions are designed to prevent IC destruction under fault conditions described in the data sheet. Fault conditions are considered as "outside" normal operating range. Protection functions are not designed for continuous repetitive operation.

[2] see also $V_{ON(CL)}$ in circuit diagram on page 8

[3] Requires a 150 Ω resistor in GND connection. The reverse load current through the intrinsic drain-source diode has to be limited by the connected load. Power dissipation is higher compared to normal operating conditions due to the voltage drop across the drain-source diode. The temperature protection is not active during reverse current operation! Input current has to be limited (see max. ratings page 3).

Electrical Characteristics

Parameter at T_j = -40...+150°C, V_{bb} = 13,5V, unless otherwise specified	Symbol	Values			Unit
		min.	typ.	max.	
Input and Status feedback					
Input turn-on threshold voltage	$V_{IN(T+)}$	-	-	2.2	V
Input turn-off threshold voltage	$V_{IN(T-)}$	0.8	-	-	
Input threshold hysteresis	$\Delta V_{IN(T)}$	-	0.3	-	
Off state input current V_{IN} = 0.7 V	$I_{IN(off)}$	1	-	30	µA
On state input current V_{IN} = 5 V	$I_{IN(on)}$	1	-	30	
Status output (open drain), Zener limit voltage I_{ST} = 1.6 mA	$V_{ST(high)}$	5.4	6.1	-	V
Status output (open drain), ST low voltage	$V_{ST(low)}$				
T_j = -40...+25 °C, I_{ST} = 1.6 mA		-	-	0.4	
T_j = 150 °C, I_{ST} = 1.6 mA		-	-	0.6	
Status invalid after positive input slope [1)]	$t_{d(ST)}$	-	300	600	µs
Input resistance (see page 8)	R_I	1.5	3.5	5	kΩ

Diagnostic Characteristics

Short circuit detection voltage	$V_{OUT(SC)}$	-	2.8	-	V
Open load detection voltage	$V_{OUT(OL)}$	-	3	-	
Openload detection current included in standby current $I_{bb(off)}$	$I_{L(OL)}$	-	5	-	µA

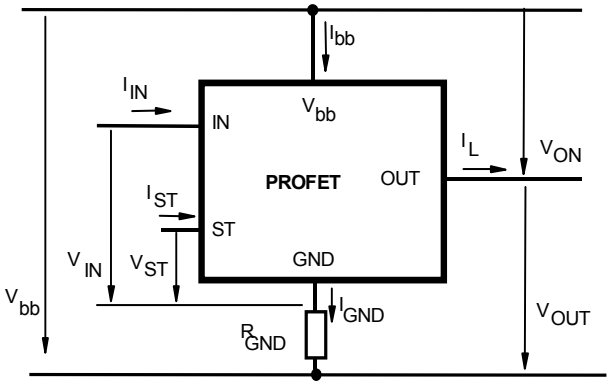
[1] no delay time after overtemperature switch off and short circuit in on-state

	Input level	Output level	Status
Normal operation	L H	L H	H H
Short circuit to GND	L H	L L *	H L
Short circuit to V_{bb} (in off-state)	L H	H H	L H
Overload	L H	L H **	H H
Overtemperature	L H	L L	H L
Open Load in off-state	L H	H H	L H

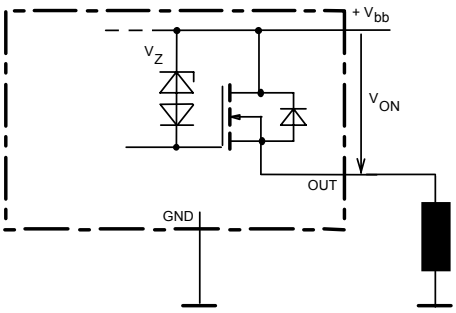
*) Out ="L": V_{OUT} < 2V typ.

**) Out ="H": V_{OUT} > 2V typ.

Terms

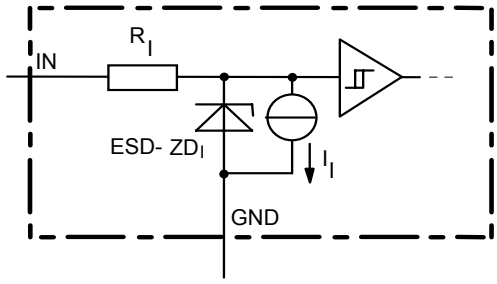


Inductive and overvoltage output clamp



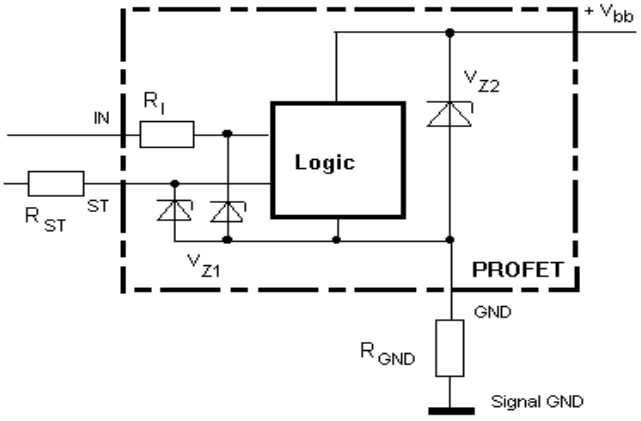
V_{ON} clamped to 47 V typ.

Input circuit (ESD protection)



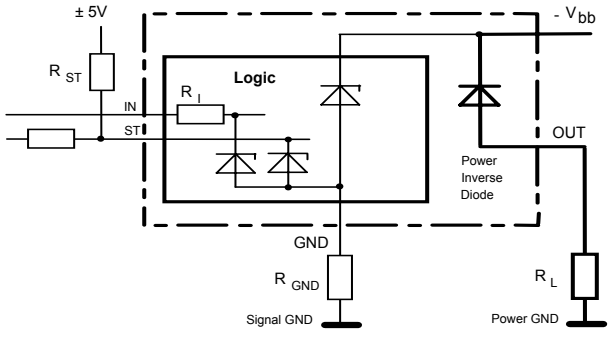
The use of ESD zener diodes as voltage clamp at DC conditions is not recommended

Overvoltage protection of logic part



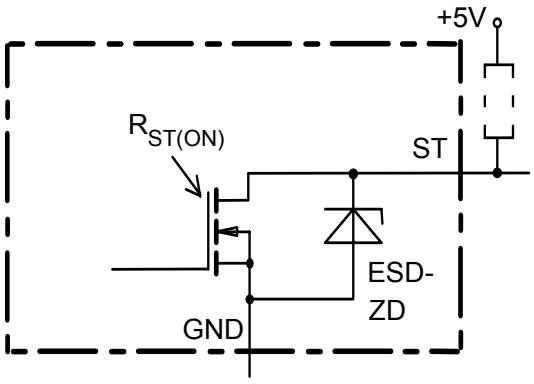
V_{Z1}=6.1V typ., V_{Z2}=$V_{bb(AZ)}$=47V typ.,
R_I=3.5 kΩ typ., R_{GND}=150Ω

Reverse battery protection



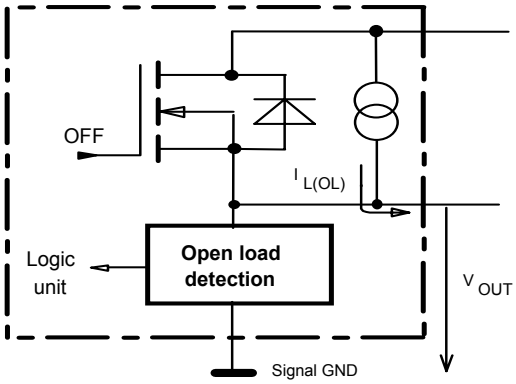
R_{GND}=150Ω, R_I=3.5kΩ typ.,
Temperature protection is not active during inverse current

Status output

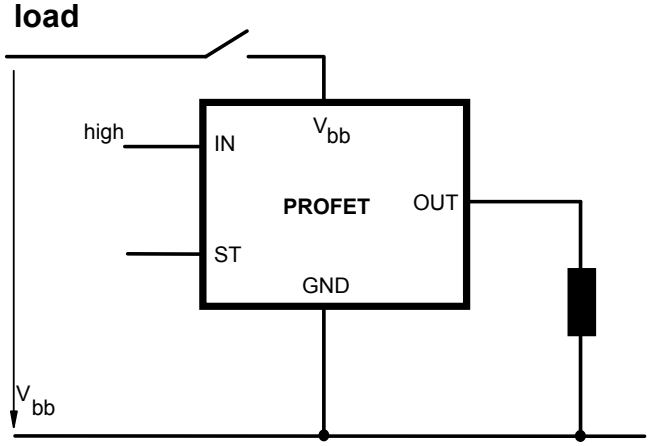


Open-load detection

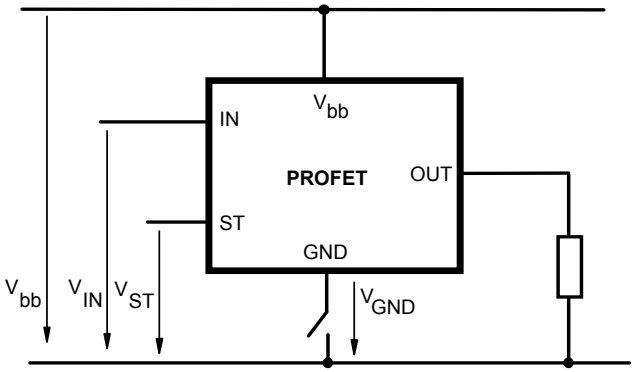
OFF-state diagnostic condition:
V_{OUT} > 3V typ.; IN=low



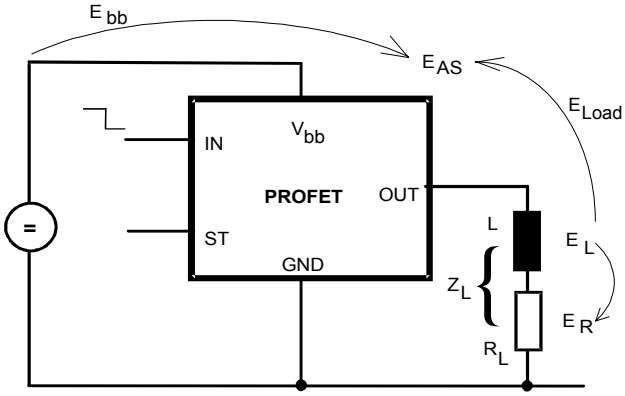
V_{bb} disconnect with charged inductive load



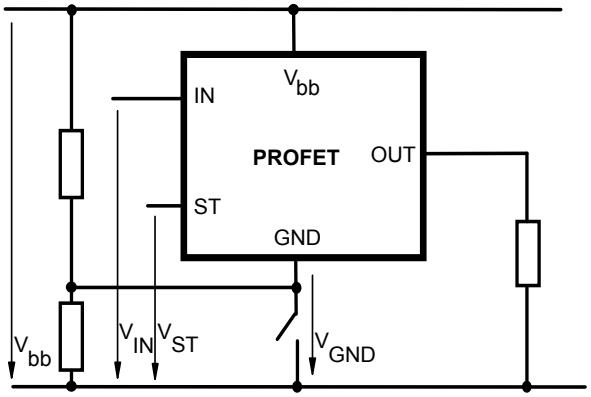
GND disconnect



Inductive Load switch-off energy dissipation



GND disconnect with GND pull up



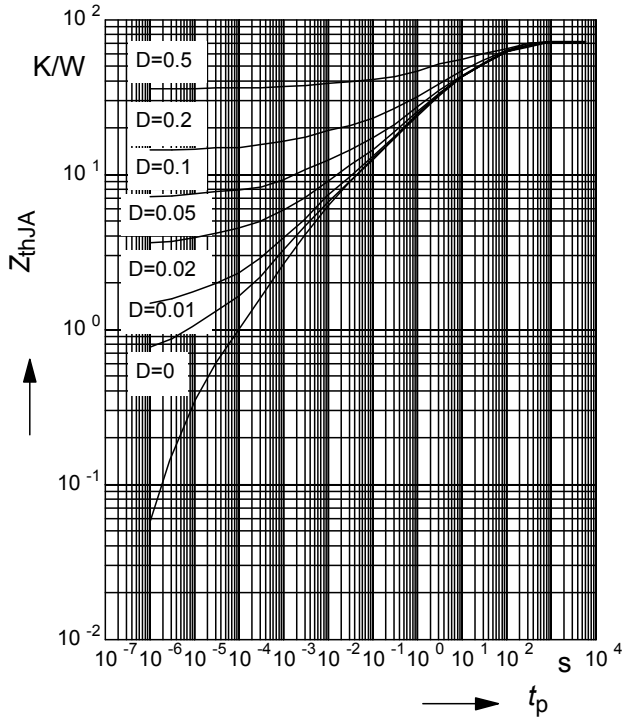
Energy stored in load inductance: $E_L = ½ * L * I_L^2$
While demagnetizing load inductance,
the enérgy dissipated in PROFET is
$E_{AS} = E_{bb} + E_L - E_R = V_{ON(CL)} * i_L(t)\,dt$,
with an approximate solution for $R_L > 0\Omega$:

$$E_{AS} = \frac{I_L * L}{2 * R_L} * (V_{bb} + |V_{OUT(CL)}|) * \ln\left(1 + \frac{I_L * R_L}{|V_{OUT(CL)}|}\right)$$

Typ. transient thermal impedance

$Z_{thJA}=f(t_p)$ @ 6cm² heatsink area

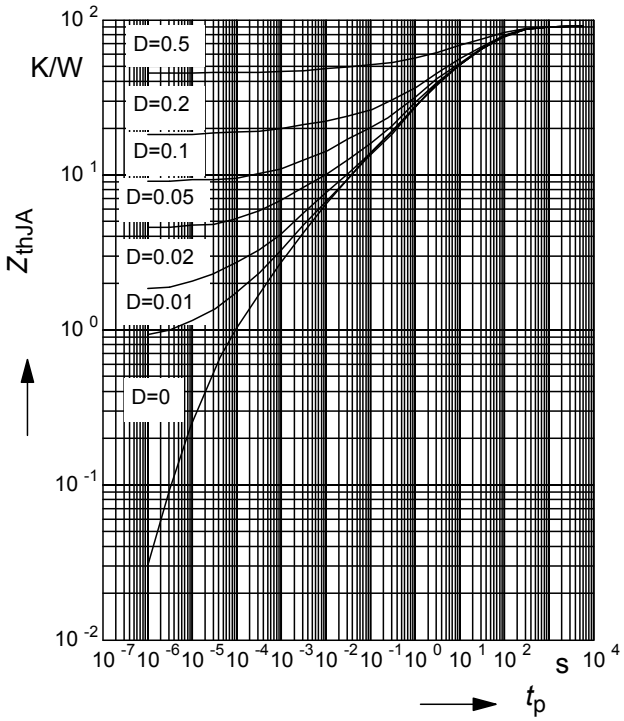
Parameter: $D=t_p/T$



Typ. transient thermal impedance

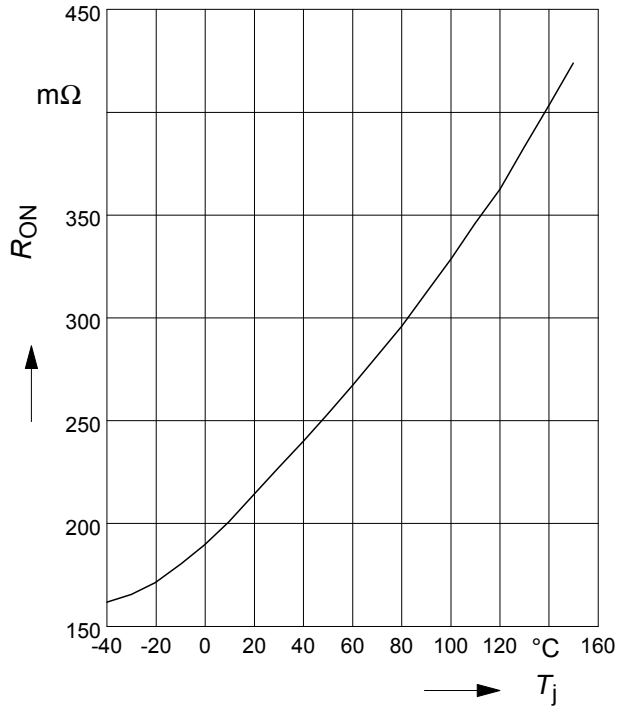
$Z_{thJA}=f(t_p)$ @ minimal footprint

Parameter: $D=t_p/T$



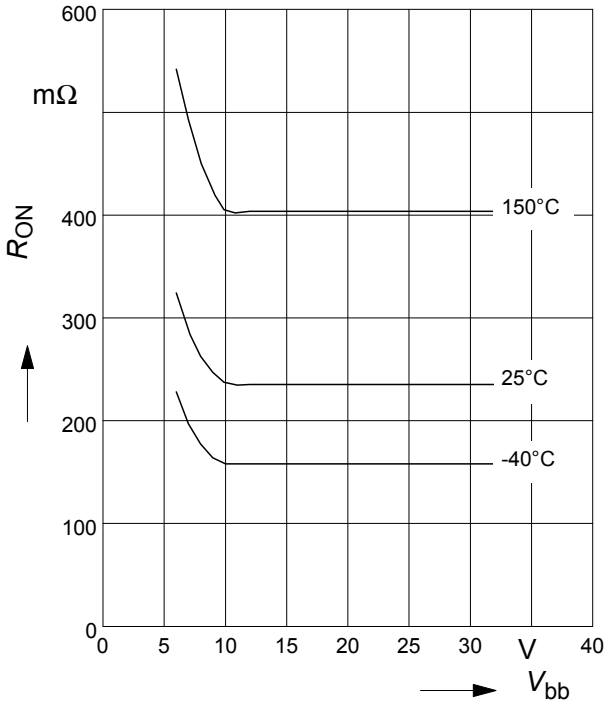
Typ. on-state resistance

R_{ON} = f(T_j) ; V_{bb} = 13,5V ; V_{in} = high



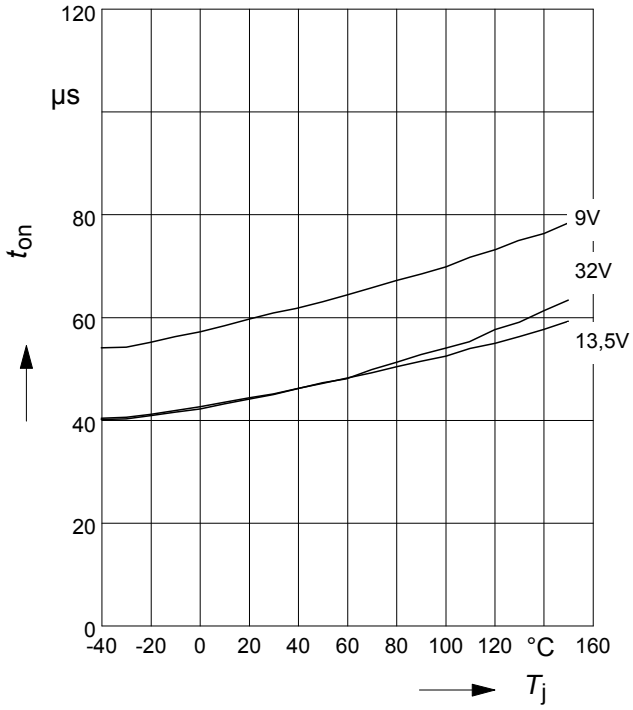
Typ. on-state resistance

R_{ON} = f(V_{bb}); I_L = 0.3A ; V_{in} = high



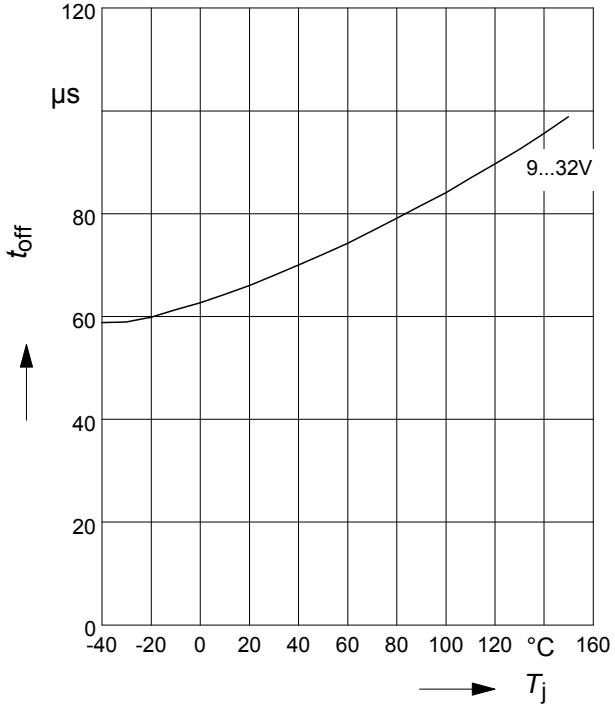
Typ. turn on time

$t_{on} = f(T_j)$; $R_L = 47\Omega$



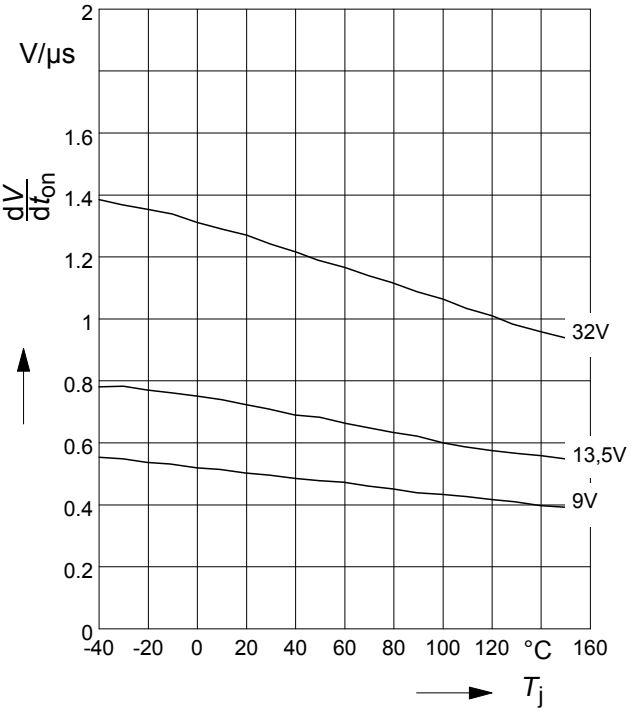
Typ. turn off time

$t_{off} = f(T_j)$; $R_L = 47\Omega$



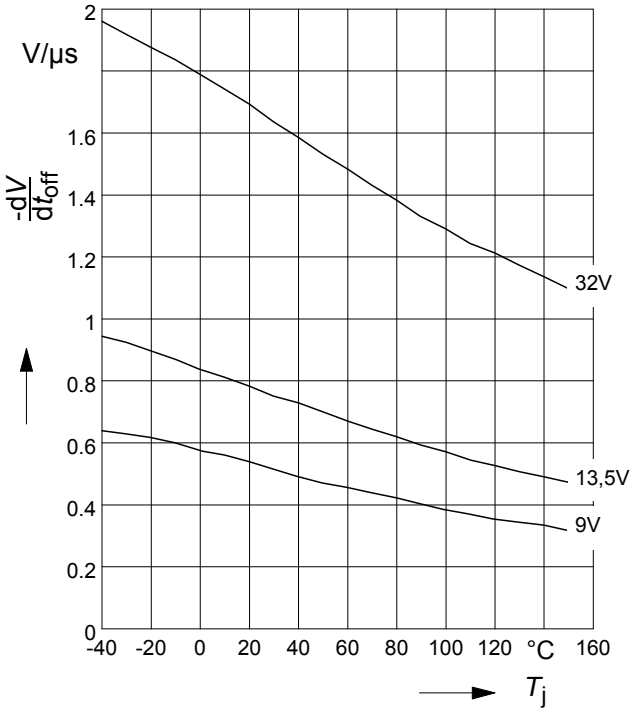
Typ. slew rate on

$dV/dt_{on} = f(T_j)$; $R_L = 47\ \Omega$



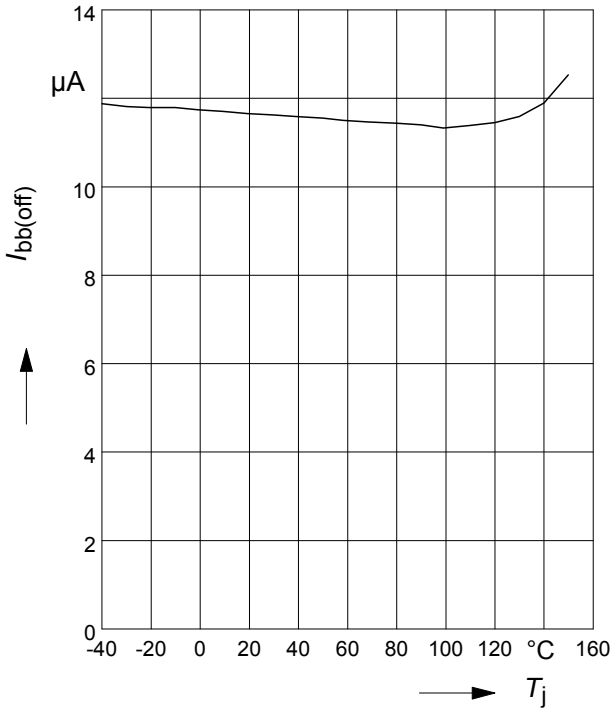
Typ. slew rate off

$dV/dt_{off} = f(T_j)$; $R_L = 47\ \Omega$



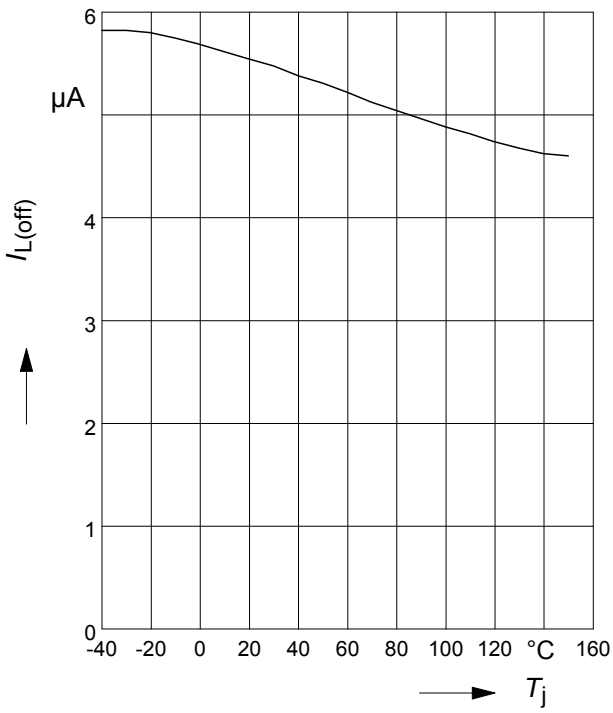
Typ. standby current

$I_{bb(off)} = f(T_j)$; V_{bb} = 32V ; V_{IN} = low



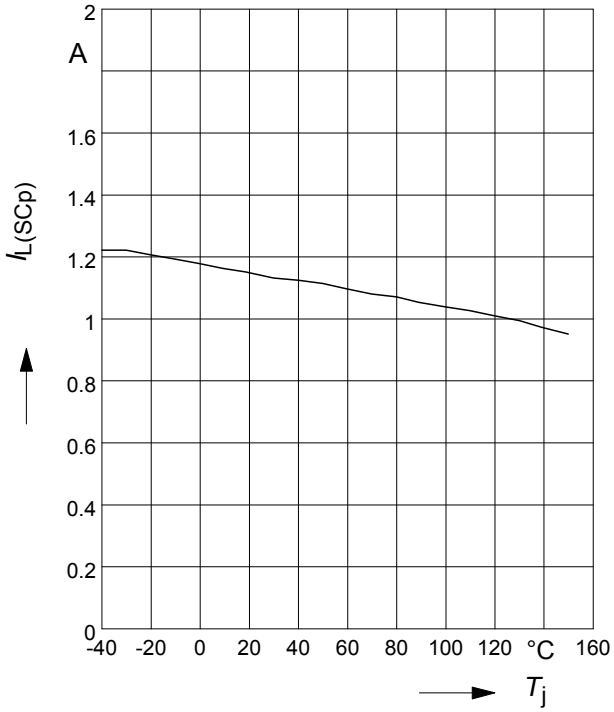
Typ. leakage current

$I_{L(off)} = f(T_j)$; V_{bb} = 32V ; V_{IN} = low



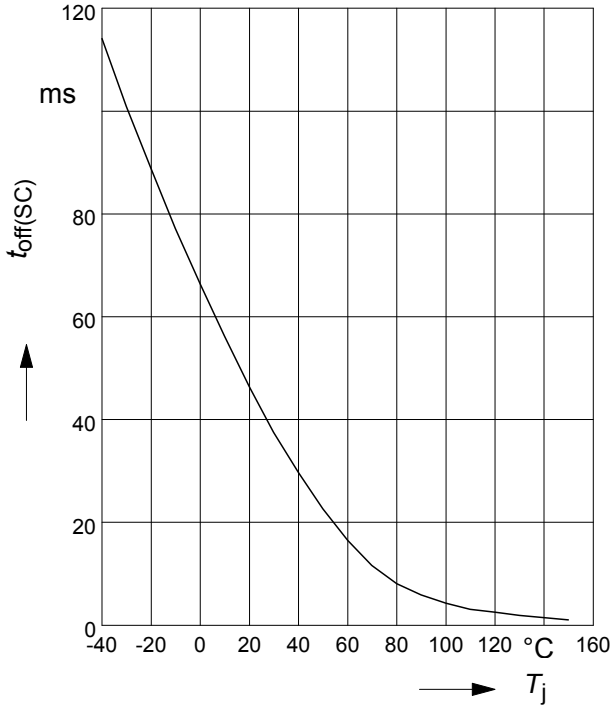
Typ. initial peak short circuit current limit

$I_{L(SCp)} = f(T_j)$; V_{bb} = 20V



Typ. initial short circuit shutdown time

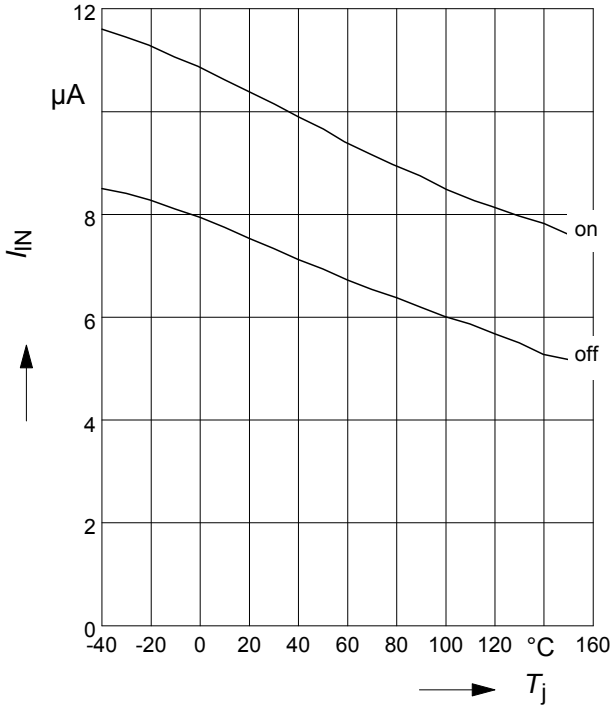
$t_{off(SC)} = f(T_{j,start})$; V_{bb} = 20V



Typ. input current

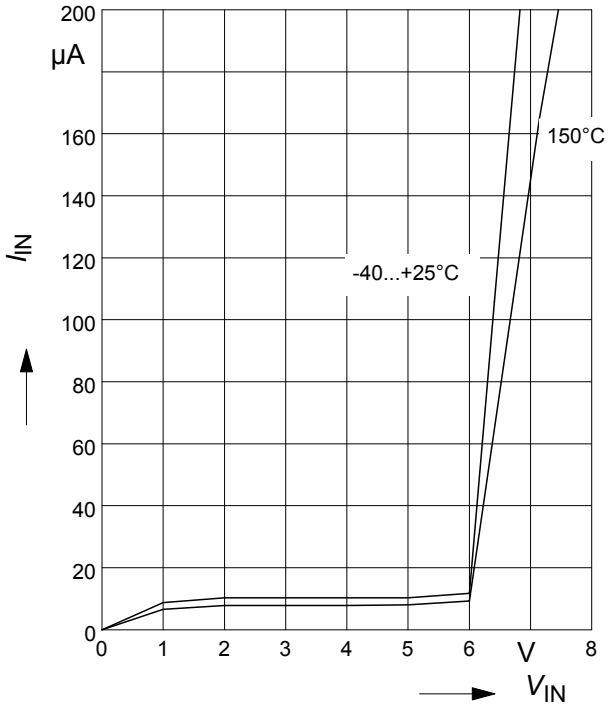
$I_{IN(on/off)} = f(T_j)$; V_{bb} = 13,5V; V_{IN} = low/high

$V_{INlow} \leq 0{,}7V$; V_{INhigh} = 5V



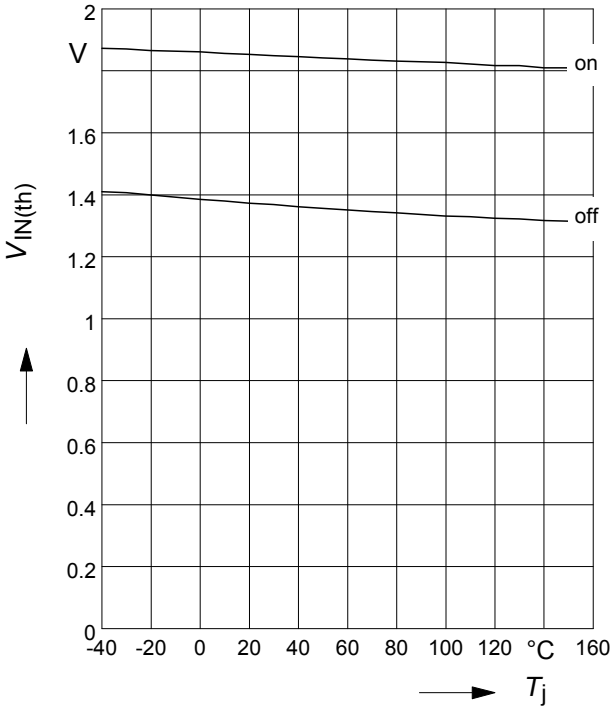
Typ. input current

$I_{IN} = f(V_{IN})$; V_{bb} = 13.5V



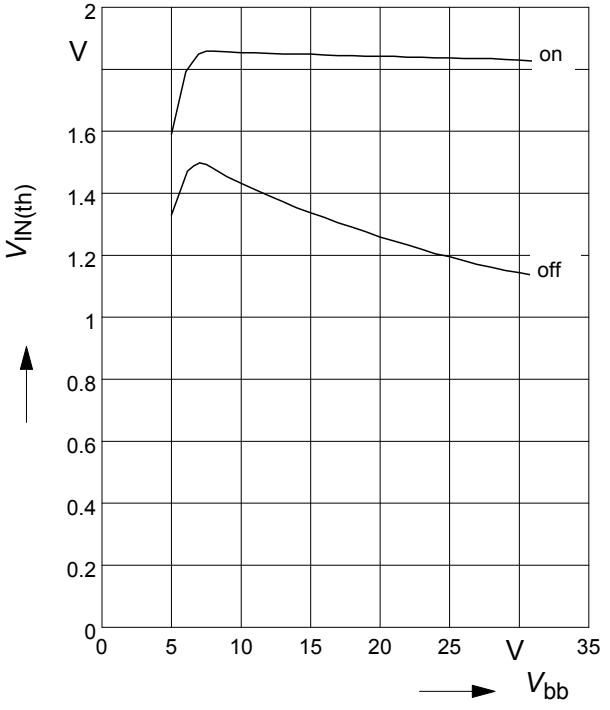
Typ. input threshold voltage

$V_{IN(th)} = f(T_j)$; V_{bb} = 13,5V



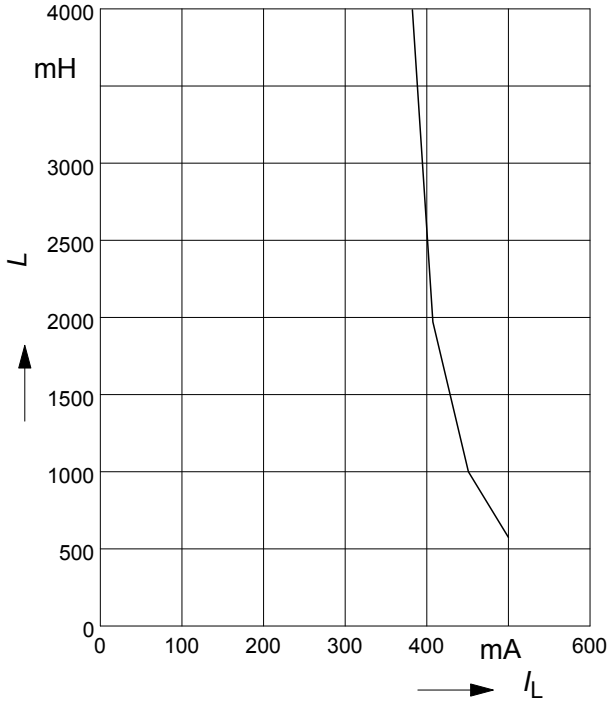
Typ. input threshold voltage

$V_{IN(th)} = f(V_{bb})$; T_j = 25°C



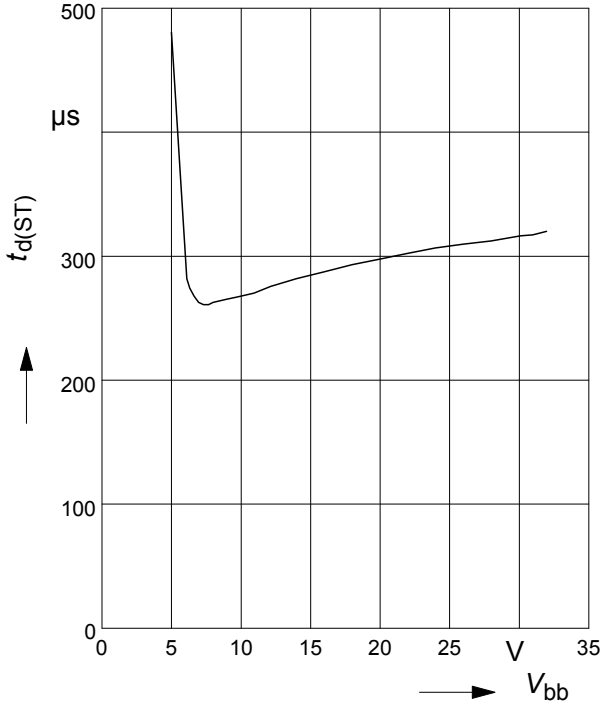
Maximum allowable load inductance for a single switch off

***L* = f(*I*L)**; T_{jstart}=150°C, V_{bb}=13.5V, R_L=0Ω



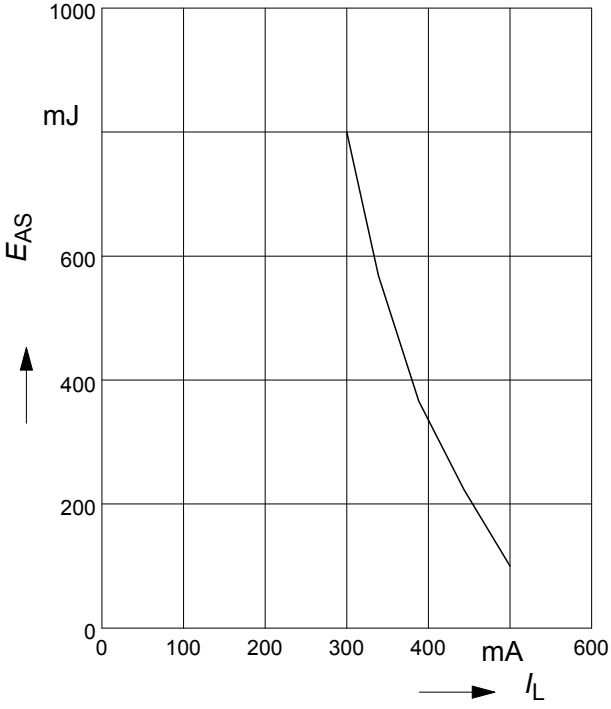
Typ. status delay time

***t*d(ST) = f(*V*bb)**; T_j = 25°C



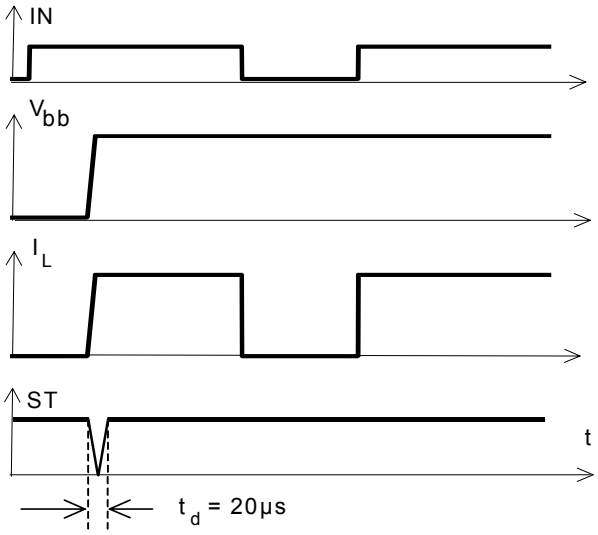
Maximum allowable inductive switch-off energy, single pulse

***E*AS = f(*I*L)**; T_{jstart} = 150°C, V_{bb} = 13,5V



Timing diagrams

Figure 1a: Vbb turn on:



Invalid status during t_d

Figure 2b: Switching a lamp,

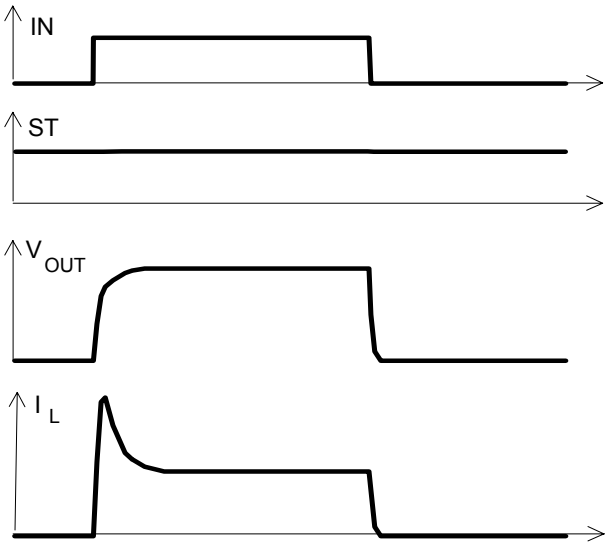


Figure 2a: Switching a resistive load, turn-on/off time and slew rate definition

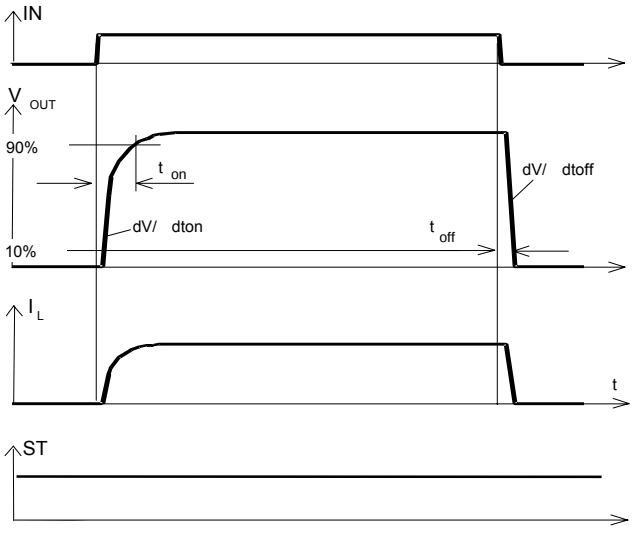


Figure 2c: Switching an inductive load

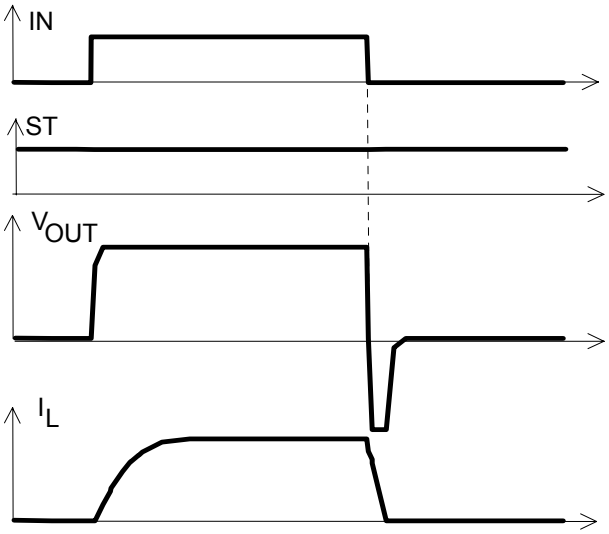
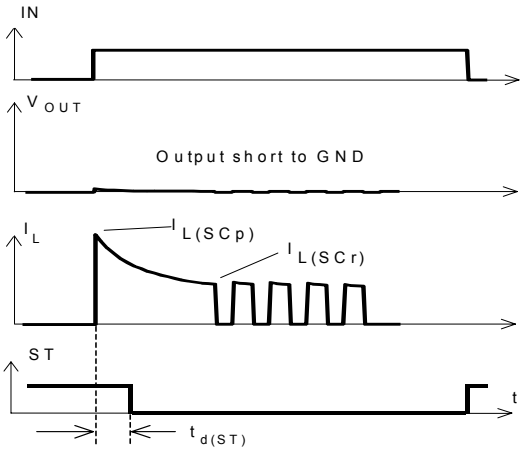


Figure 3a: Turn on into short circuit,
shut down by overtemperature, restart by cooling



Heating up of the chip may require several milliseconds, depending on external conditions.

Figure 4: Overtemperature:
Reset if $T_j < T_{jt}$

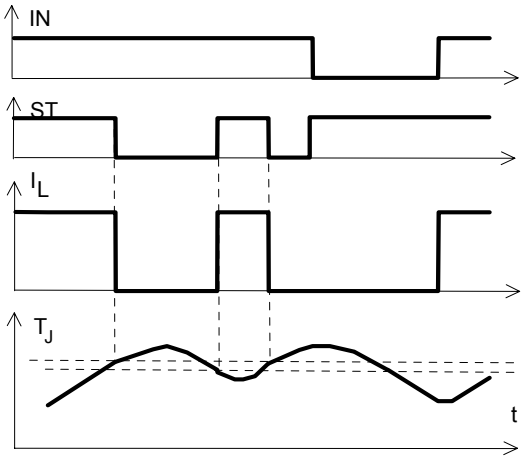


Figure 3b: Short circuit in on-state
shut down by overtemperature, restart by cooling

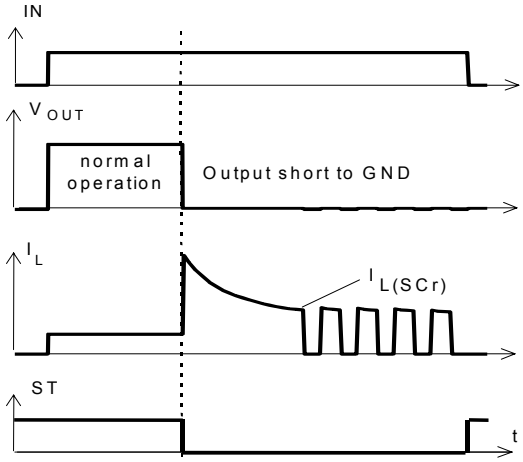
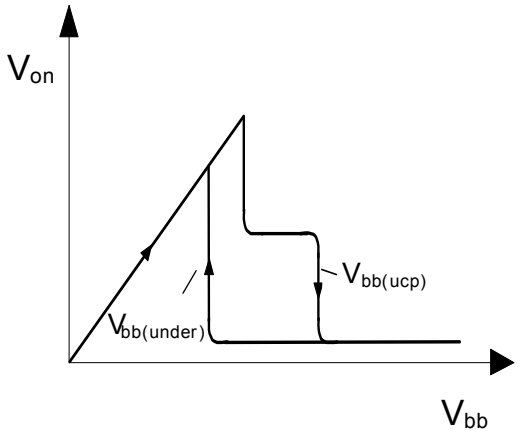


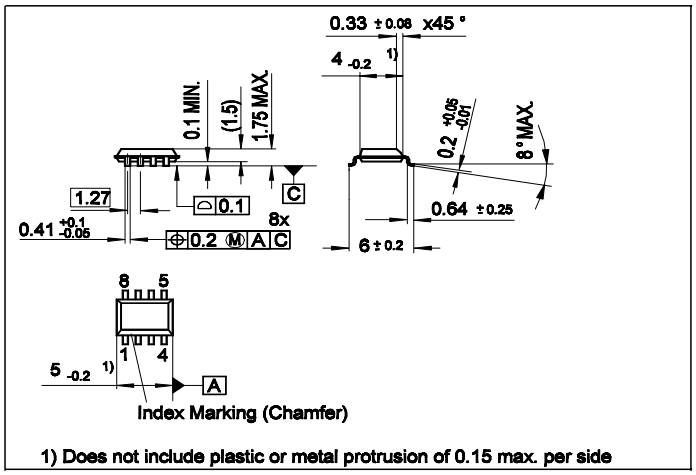
Figure 5: Undervoltage restart of charge pump

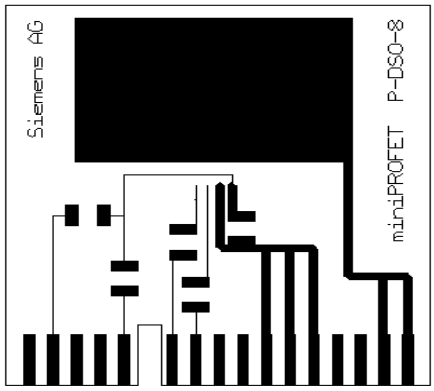


Package and ordering code

all dimensions in mm

Package:	Ordering code:
P-DSO-8-6	Q67060-S7303-A2





Printed circuit board (FR4, 1.5mm thick, one layer 70µm, 6cm^2 active heatsink area) as a reference for max. power dissipation P_{tot} nominal load current $I_{L(nom)}$ and thermal resistance R_{thja}

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